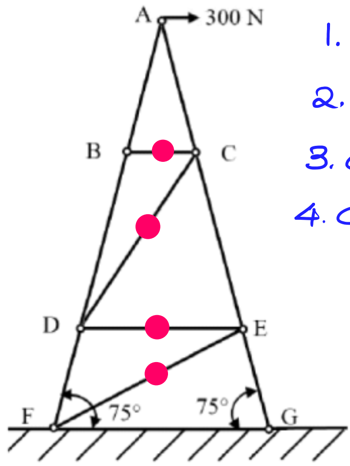
1 Mark

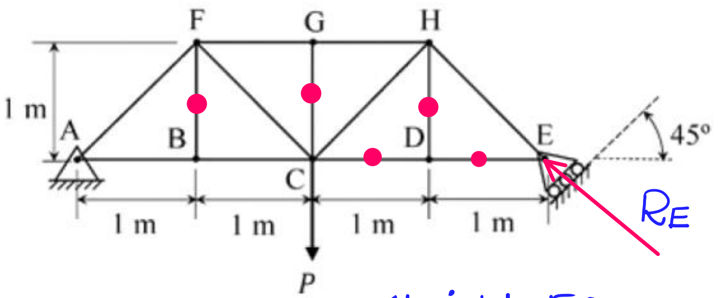
The figure shows an idealized plane truss. If a horizontal force of 300 N is applied at point A, then the magnitude of the force produced in member CD is ___0___ N.

Identify Number of zero force Members. {4}



1. observe joint B
2. observe joint C
3. observe joint D
4. observe joint E

Q. The members carrying zero force (i.e. zero-force members) in the truss shown in the figure, for any load $P > 0$ with no appreciable deformation of the truss (i.e. with no appreciable change in angles between the members), are



(A) BF and DH only

(B) BF, DH and GC only

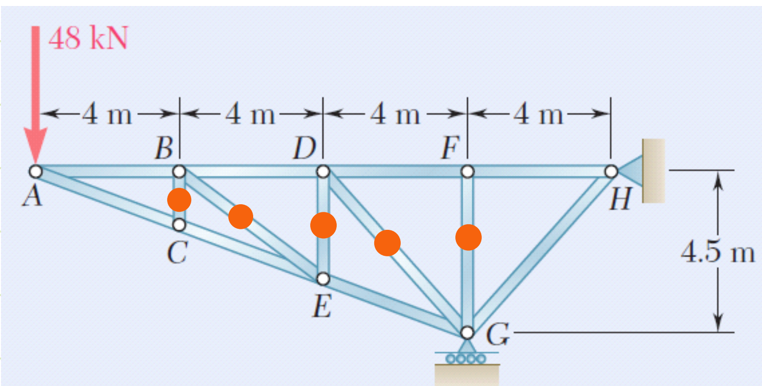
(C) BF, DH, GC, CD and DE only ✓

(D) BF, DH, GC, FG and GH only

At joint E:

R_E & EH are Collinear

∴ DE is a zero force Member.



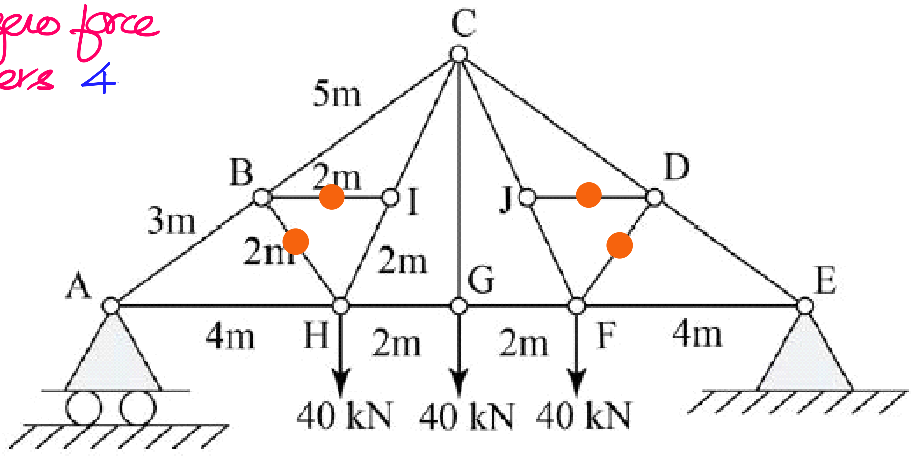
Identify the Number of Zero force Members.

5

1 Mark

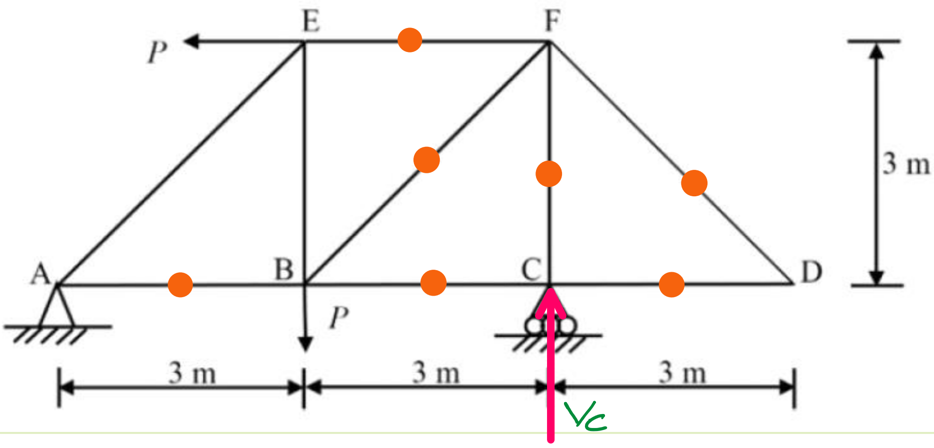
Find the force (in kN) in the member BH of the truss shown.

No. of zero force Members 4



2 Mark

A pin-jointed truss has a pin support at A and a roller support at C. All the members are made of same material and have the same cross-section. Neglect the self-weight of the members. Due to the applied loading shown, the total number of zero force members is 7.



$V_c = ?$; $\Sigma M_A = 0$ for the entire structure.

$P\{3\} - P\{3\} + V_c\{6\} = 0$

$\therefore V_c = 0$

$T_{CF} = V_c$ $\quad T_{CF} = 0$

Observe joint F after removing CF $\quad T_{BF} = T_{EF} = 0$

observe joint B after removing BC & BF $\quad T_{AB} = 0$

Analysis of Frames

Frames are structures in which at least one member exp. more than 2 forces leading to Bending effect apart from Axial loading.

Analysis of Frames:

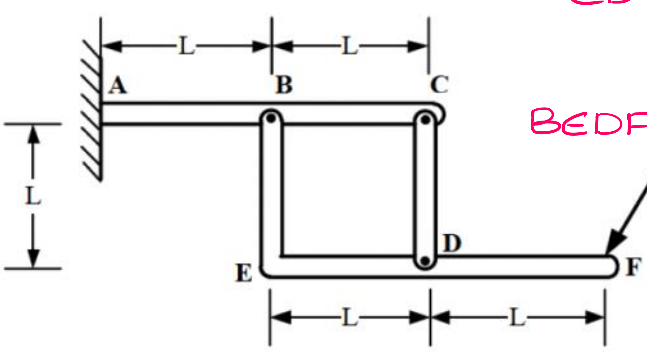
Step 1: Draw FBD of entire structure as a single body without dis-assembling the Members {∴ the loadings in the Members behave as Internal loadings.}

Step 2: Establish eq^m by considering the applied loads & the support Reac^ns {Make use of 2 force Members' or 3 force Members' logics to identify dir^ns of support Reac^ns if those members are present in the system}.

Step 3: If the loadings in each member have to be found, then dis-assemble entire structure, draw FBD & establish eq^m for each Member.

1 Mark

The frame comprises members ABC, CD, and BEDF. This frame is fixed at A, and the connections between the three members are pins. Load P is applied at F. Which of the following statements is TRUE?

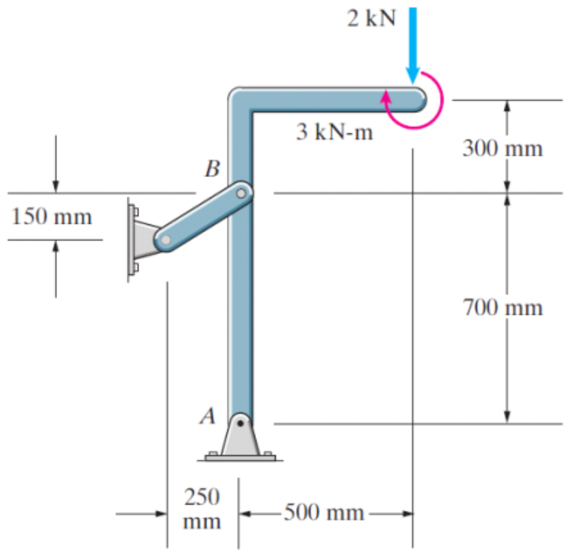


CD: forces at C & D.
∴ CD is a 2 force Member. {axial}

BEDF: forces at B, D & F
∴ BEDF is a 3 force Member {conc.}

ABC: forces at A, B & C, but it can't be regarded as 3 force Member since there is a Moment at A.

(A) ABC and CD are two-force members
(B) CD is the only two-force member
(C) BEDF and CD are two-force members
(D) ABC is the only two-force member



(a) Is the L-shaped bar a three-force member?

(b) Are the three forces acting on the L-shaped bar concurrent?

(c) Determine the magnitudes of the reactions at A and B.

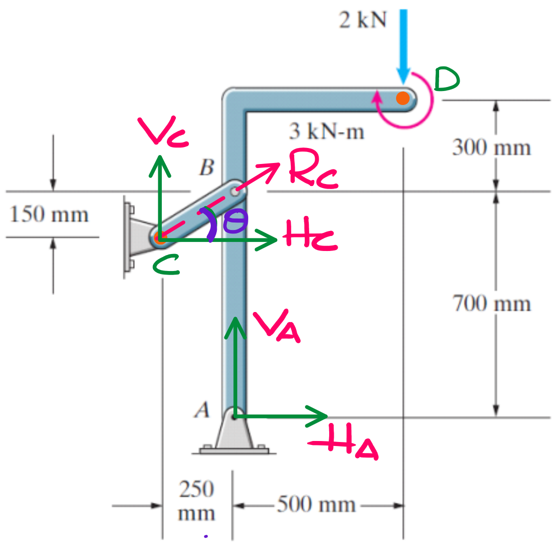
a. Even though 3 forces $\{R_A, R_B, 2\,kN\}$ act on the L shaped bar, still it can't be regarded as a 3 force Member due to $M = 3\,kN.m$.

b. Let us assume the 3 forces $\{R_A, R_B, 2\,kN\}$ are conc at "O"

$\therefore$ for the L shaped bar $\Sigma M_O = 0$

$$\Rightarrow \cancel{M_O^{R_A}} + \cancel{M_O^{R_B}} + \cancel{M_O^{2kN}} - 3\,kN.m = 0 \Rightarrow 3\,kN.m = 0$$

$\therefore$ Our assumption is wrong.



$\because$ BC is a 2 force Member {axial}

$\therefore$ the net reac^n of H_c & V_c passes thru B & also V_A passes thru B.

$\therefore \Sigma M_B = 0$

$$-3\,kN.m - 2\,kN\{0.5\} + H_A\{0.7\} = 0$$

$$\boxed{H_A = 5.714\,kN.}$$

$\Sigma F_x = 0$

$H_A + H_c = 0$ $\quad \boxed{H_c = -5.714\,kN.}$

$$\tan\theta = \frac{150}{250} = \frac{V_c}{H_c}$$

$$\boxed{V_c = -3.428\,kN.}$$

Axial loading in 2 force Member.

$\Sigma F_y = 0$

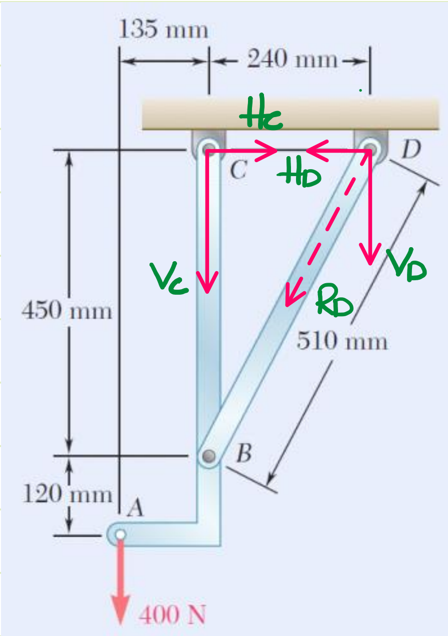
$V_c + V_A - 2 = 0$

$V_A = 2 - V_c$

$$\boxed{V_A = 5.428\,kN.}$$

Q. In the following cases identify 2 force & 3 force Members & also find support R_x^{ns}.

2 Marks



BD : 2 force Member {axial}

ABC : 3 force Member {concurrent}

∵ the resultant of H_D & V_D passes thru B & also V_C passes thru B.

$\Sigma M_B = 0 \Rightarrow 400\{0.135\} - H_C\{0.45\} = 0$

$\boxed{H_C = 120\,N}$

$\Sigma F_x = 0 \Rightarrow H_C - H_D = 0$ $\quad \boxed{H_D = 120\,N}$

$\Sigma M_D = 0 \Rightarrow 400\{375\} + 240 V_C = 0$

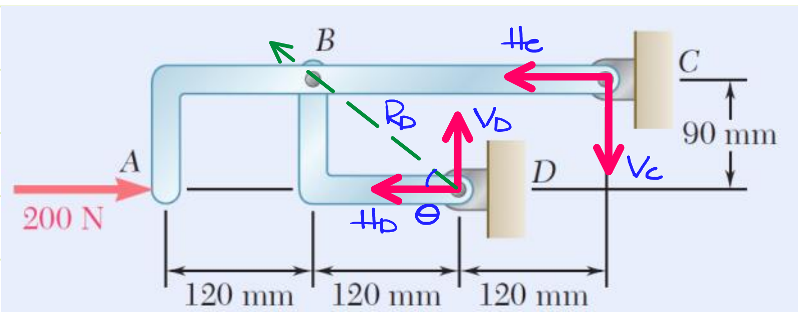
$\boxed{V_C = -625\,N}$

$\Sigma F_y = 0 \Rightarrow -400 - V_C - V_D = 0$

$V_D = -V_C - 400$

$\boxed{V_D = 225\,N}$

2. Marks



ABC : 3 force M. (conc.)

BD : 2 force M (along BD)

∵ the resultant of V_D & H_D pass thru B & H_C also passes thru B.

$\Sigma M_B = 0$

$200\{90\} - V_C\{240\} = 0$ $\quad \boxed{V_C = 75\,N.}$

$\Sigma F_y = 0$

$V_D - V_C = 0$ $\quad \boxed{V_D = 75\,N}$

$\tan\theta = \frac{90}{120} = \frac{V_D}{H_D} \Rightarrow H_D = \frac{75 \times 120}{90}$ $\quad \boxed{H_D = 100\,N.}$

$\Sigma F_x = 0 : 200 - H_C - H_D = 0$ $\quad \boxed{H_C = 100\,N.}$

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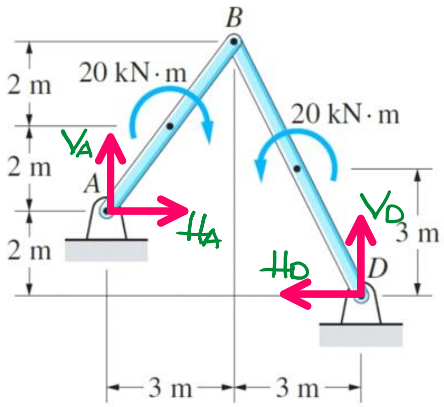
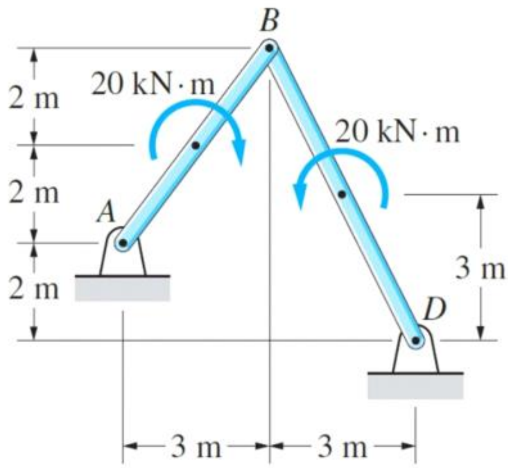
2.Mark Find the support Reac^{ns}





For the entire structure :

$\Sigma F_x = 0:$

$$\boxed{H_A = H_D}$$

$\Sigma F_y = 0:$

$$\boxed{V_A = -V_D}$$

$\Sigma M_A = 0: \; 2H_D - 6V_D = 0$

$$\Rightarrow \boxed{H_D = 3V_D}$$

∵ We have no further eq^{ns} to analyse the entire system

∴ We must examine individual Members.

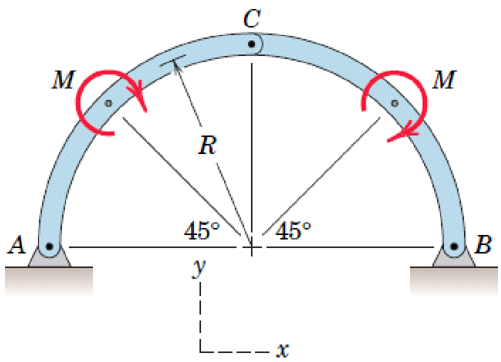
For Member BD : $\Sigma M_B = 0$

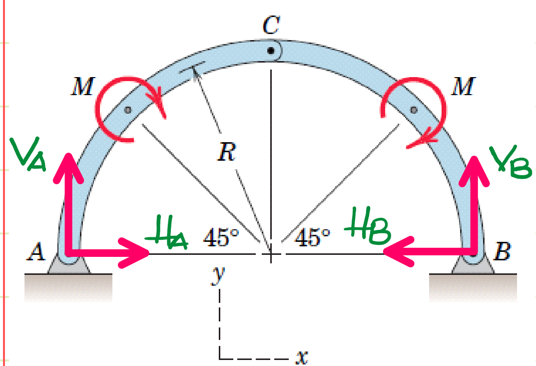
$20 - H_D\{6\} + V_D\{3\} = 0 \Rightarrow 15V_D = 20$

$$\boxed{H_D = H_A = 4\,kN \qquad V_D = \frac{4}{3}\,kN \qquad V_A = -\frac{4}{3}\,kN.}$$

2 Mar

* Find the support Reac^{ns}.





for the entire structure :

$\Sigma F_x = 0:$

$$\boxed{H_A = H_B}$$

$\Sigma F_y = 0: \; \boxed{V_A = -V_B}$

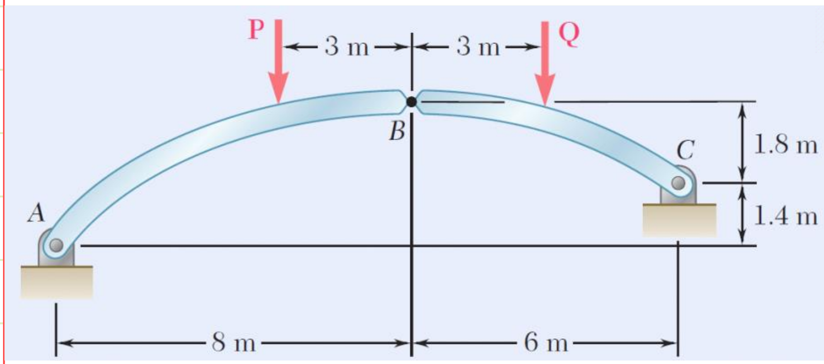
$\Sigma M_A = 0: \; -M - M + V_B\{2R\} = 0$

$$\Rightarrow \boxed{V_B = \frac{M}{R}} \qquad \boxed{V_A = \frac{-M}{R}}$$

for the Member AC :

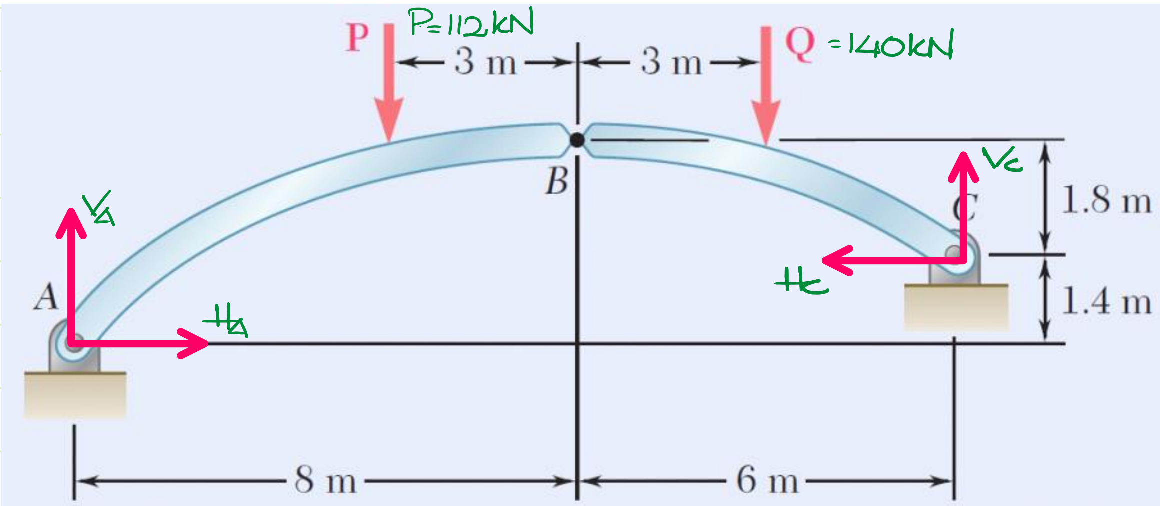
$\Sigma M_C = 0 \Rightarrow -M + H_A(R) - V_A(R) = 0$

$\Rightarrow -M + H_A(R) + \frac{M}{R}(R) = 0 \Rightarrow \boxed{H_A = H_B = 0}$



$P = 112$ kN and $Q = 140$ kN.

Find the support reac^{ns} at A & B.



$\Sigma F_y = 0$ $\quad \boxed{V_A + V_c = 252 \text{ kN.}}$

$\Sigma M_A = 0 \Rightarrow -112(5) - 140(11) + 1.4 H_c + 14 V_c = 0$

$1.4 H_c + 14 V_c = 2100$

$\boxed{H_c + 10 V_c = 1500}$

for the Member BC:

$\Sigma M_B = 0 \Rightarrow -140(3) - 1.8(H_c) + 6 V_c = 0$

$\boxed{-H_c + \frac{10}{3} V_c = \frac{700}{3}}$

$\frac{40}{3} V_c = \frac{5200}{3} \Rightarrow$ $\boxed{\begin{array}{l} V_c = 130 \text{ kN} \\ H_c = 200 \text{ kN} \\ H_A = 200 \text{ kN} \\ V_A = 122 \text{ kN.} \end{array}}$

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2.3. Friction

* Introduction
* Laws of Dry Friction
  * Case I : Particle Body
  * Case II : Rigid Body
  * Case III : Rolling Resistance
* Terminologies
  * Angle of friction {ϕ}
  * Cone of friction
  * Angle of Repose {ψ}
* Standard Problems
  * Mating Block
  * Belt friction
  * Wedge friction
  * Screw jack

Friction : The resistance against relative motion {or} tendency of relative motion b/w 2 bodies in contact is known as friction.

A. Dry friction {Columbs' friction} solid-solid contact

B. Wet friction {Viscous / Lubricating fr.} solid-fluid contact.

* Laws of Dry Friction :

1. opposes relative motion {or} tendency of Relative Motion.
2. independent of area of contact b/w the two bodies.
3. depends on the Normal ac^n-reaction pair b/w the two bodies in contact
4. Static friction : The frictional force existing b/w two bodies which are at relative rest. It is a self adjusting force which tries to oppose tendency of relative motion.

   Its maximum value, Limiting static friction {f_s} is attained when the bodies impend to have relative motion.

   $\boxed{f_s = \mu_s . N}$ μ_s : Coeff. of static friction.

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5. Kinetic / Dynamic friction: It is the friction attained when relative motion takes place. It is lower than Limiting static friction & is a constant.

Kinetic friction: $\boxed{f_k = \mu_k . N.}$ μ_k: Coeff. of kin. friction.

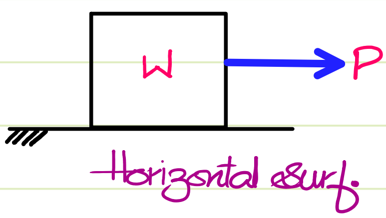
$\boxed{\because f_k < f_s} \Rightarrow \boxed{\therefore \mu_k < \mu_s.}$

6. At extremely high relative velocities kinetic friction keeps decreasing.

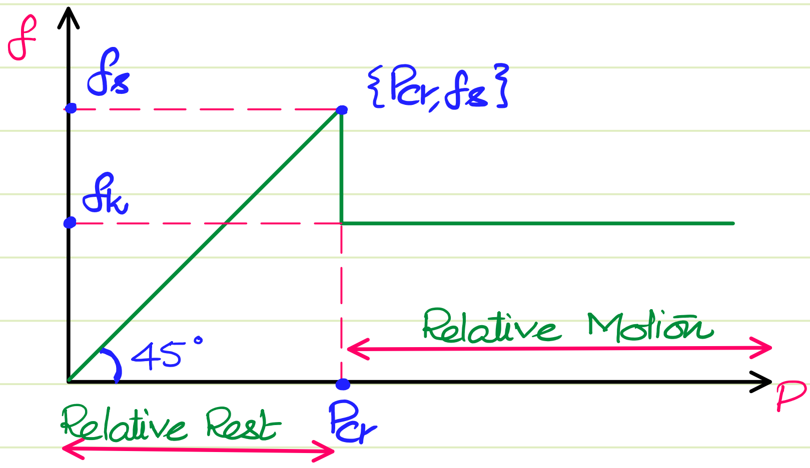
Note: Coeff. of friction are binary properties {defined for a pair of bodies in contact}

* Case I: Friction On A Particle Body:

If the dimensions of a body are not specified, then the body has become a particle body. All the forces on particle body must be concurrent.

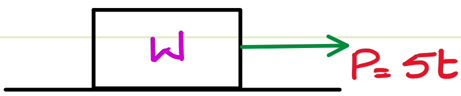


$P_{cr} = f_s = \mu_s . N.$



1. $P \leq P_{cr}$: {Relative Rest} $f = P$
2. $P > P_{cr}$: {Relative Motion} $f_k = \mu_k . N.$

Q.



$W = 100N$; $\mu_s = 0.4$; $\mu_s = 0.3$

find the values of friction at

a. $t_1 = 5s$

b. $t_2 = 11s$.

$N = 100N \Rightarrow f_s = 40N$.

$\therefore P_{cr} = 40N$.

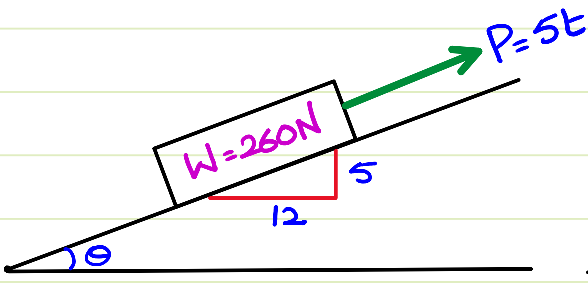
at $t_1 = 5s$: $P = 25N$. $\because P < P_{cr}$

$\therefore f = P$ $\therefore f = 25N$.

at $t_2 = 11s$: $P = 55$ $\because P > P_{cr}$

$\therefore f = f_k = \mu_k N$ $\therefore f_k = 30N$.

Q.



A force $P = 5t$ parallel to the incline acts as shown

$\mu_s = 0.5$; $\mu_k = 0.25$

find the values of friction at

$t_1 = 10s$, $t_2 = 25s$, $t_3 = 50s$.

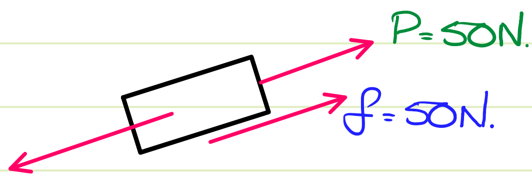
$\sin\theta = \frac{5}{13}$; $\cos\theta = \frac{12}{13}$

$N = W.\cos\theta = 240N$.

$f_s = \mu_s N = 120N$.

$f_k = \mu_k N = 60N$.

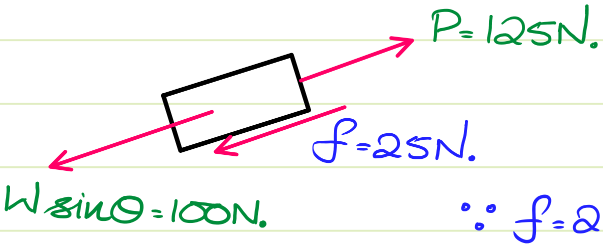
a. @ $t_1 = 10s$



$\because 50N < f_s$

$\therefore 50N$ is acceptable friction.

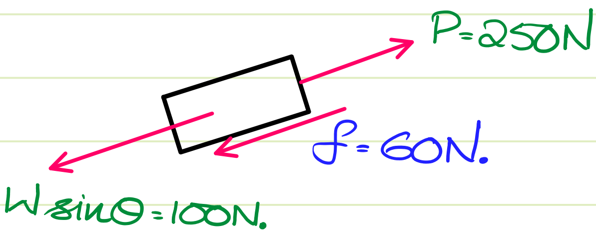
b. at $t = 25s$.



$\because f = 25N < f_s$

$\therefore$ 25N is acceptable value.

c. at $t = 50s$

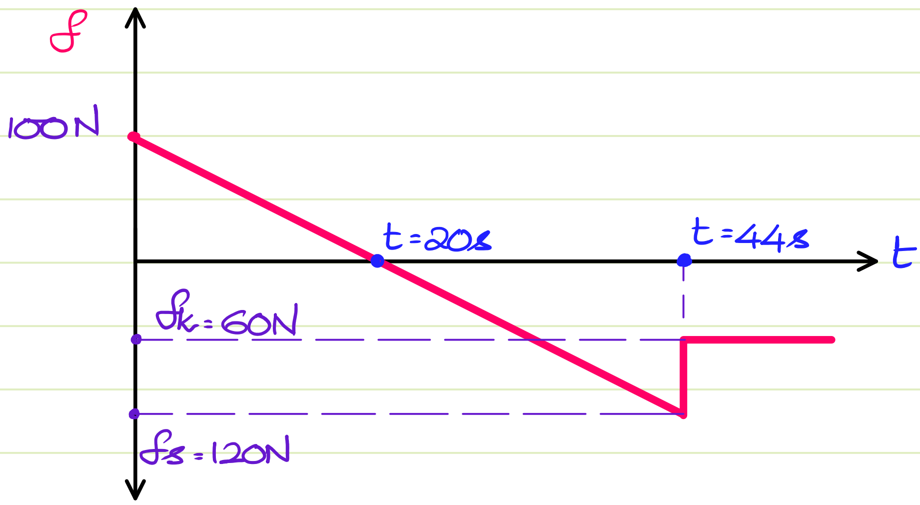


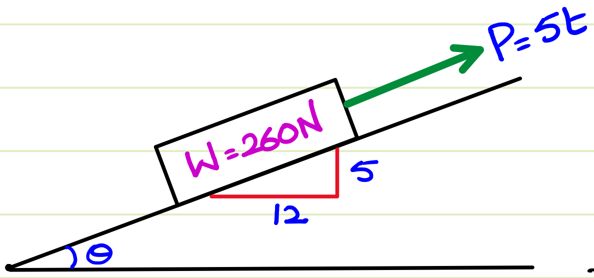
$\because$ the required $f = 150N > f_s$

$\therefore f = 150N$ is impossible

$\therefore$ the body is in dyn cond

& $f = f_k = 60N$.



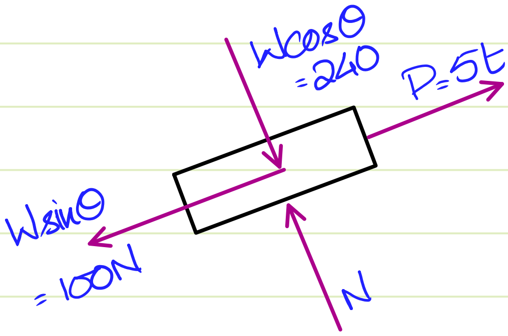


A force P=5t parallel to the incline acts as shown

$\mu_s = 0.5$; $\mu_k = 0.25$

find the values of friction at $t_1 = 10s$, $t_2 = 25s$, $t_3 = 50s$.

$\sin\theta = \frac{5}{13}$; $\cos\theta = \frac{12}{13}$



$N = 240N$

$f_s = 120N$ $\quad f_k = 60N$.

* If the body demands any amount of force less than or equal to f_s in order to remain at rest, then that value of force is acceptable for st. friction

i) At $t_1 = 10s$; $P = 50N$.

demand for eq^m : $f = 50$

ii) At $t_2 = 25s$; $P = 125N$.

demand for eq^m : $f = 25$

iii) At $t_3 = 50s$; $P = 250N$.

demand for eq^m : $f = 150$ ×

We have dyn cond^n $\therefore$ $f = f_k = 60N$

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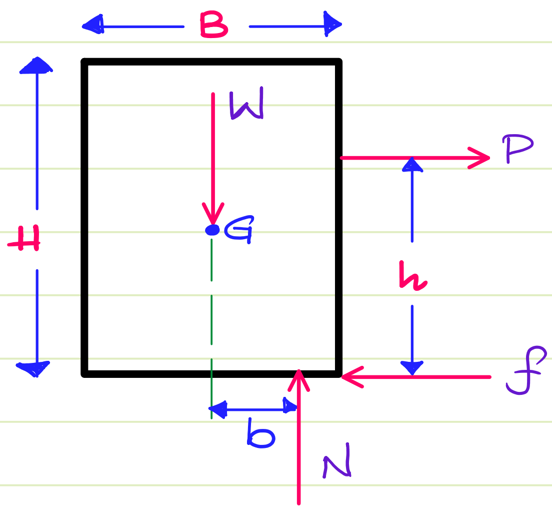
Case 2: Friction In Case of Rigid Body:

In case of rigid bodies, the forces acting may not be concurrent ⇒ There is a possibility of Moment (or) couple to be induced.

If the applied force & friction creates a disturbing couple then W & N try to create a restoring couple & for this to happen W & N must become eccentric w.r.t. each other.



$\Sigma F_x = 0$

$\Rightarrow P = f$

$\Sigma F_y = 0$

$\Rightarrow W = N$

$\Sigma M = 0$

$f h = W.b$

There are two possibilities due to these forces.

1. Toppling Effect: Normal reaches the corner about which toppling is to take place before friction could attain limiting static value.
2. Sliding Effect: Friction attains limiting static value $\{f_s\}$ before Normal could reach the point of toppling.

*** How to solve Numericals?

Step 1: Find f_s & the Max. possible eccentricity b/w W & N.

Step 2: Find these 2 couples: $C_1 = C$ by P & f_s. {Max. dist. Coup}

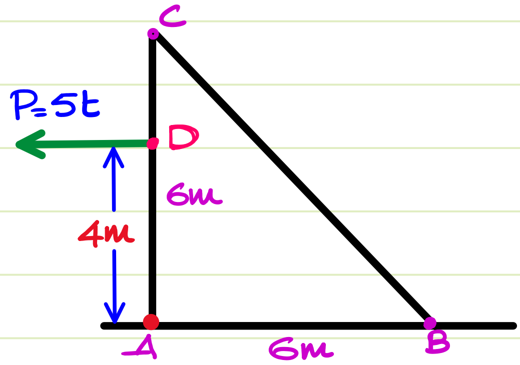
$C_2 =$ Max C by W & N {Max rest. Couple}

Step 3 : Compare C_1 & C_2

a. Couple by P & f_s < Max Couple by W & N. ⇒ {Sliding}

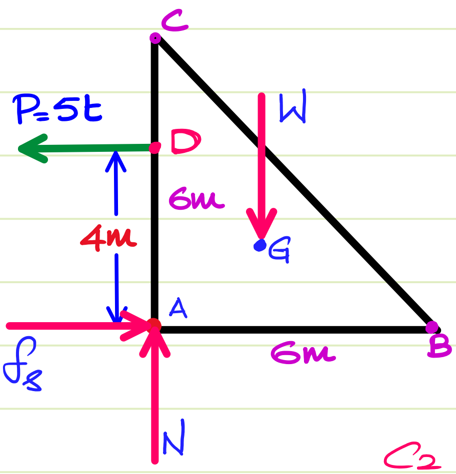
b. Max Couple by W & N < Couple by P & f_s. ⇒ {Toppling}

2 Marks.



$W = 200N$; $\mu_s = 0.6$

find the time at which Motion impends & also find the type of Motion that impends.



$f_s = \mu_s . N = 0.6 \times 200$ $\therefore f_s = 120N.$

Max possible ecc. b/w W & N = 2m.

C_1 : Max disturbing couple $= f_s \times 4$

$= 120 \times 4$

$\therefore C_1 = 480 N.m.$

C_2 : Max. restoring couple $= W \times 2$

$= 200 \times 2$

$\therefore C_2 = 400 N.m.$

$\because C_2 < C_1$ $\therefore$ Toppling takes place.

When toppling is impending to take place, eq^m is valid.

$\therefore \Sigma M_A = 0$

$W\{2\} = 5t\{4\}$

$\Rightarrow t = 400/20$

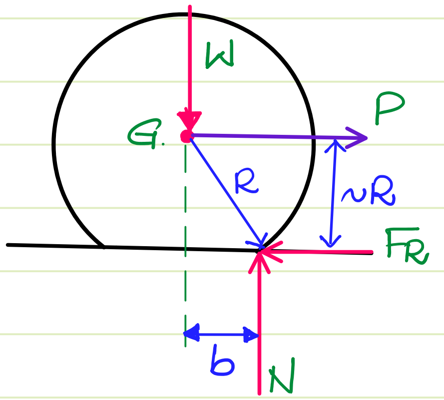
$\therefore t = 20s.$ $P = f = 5 \times 20$

$= 100N. < f_s.$

Case 3: Rolling Resistance: The resisting force acting on a body which is trying to roll but is not able to roll is known as Rolling Resistance. In the case of Rolling Resistance the bodies & surfaces are assumed to deform in order to create eccentricity b/w Weight & Normal.

There are 3 possible types of deformation.

1. Rolling body alone is deforming ***
2. Surface is deforming
3. both the rolling body & surface deform.



$\Sigma F_x = 0 \Rightarrow F_R = P$

$\Sigma F_y = 0 \Rightarrow N = W.$

$\Sigma M = 0 \Rightarrow \{F_R\}R = W.b$

$\Rightarrow F_R = \frac{W.b}{R}$

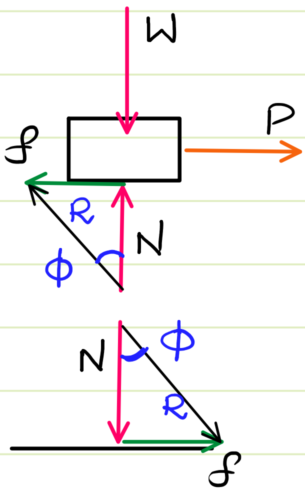
F_R : Rolling Resistance

b : Coeff. of Rolling Resistance.

* Standard Terms In Friction :

1. **Angle of friction : $\{\phi\}$** The angle made by Resultant of N & f with the Normal is Angle of friction. $\{\phi\}$

$\phi = \tan^{-1}\{f/N\}$



- Angle of static friction :
  $\phi_s = \tan^{-1}\mu_s.$
- Angle of kinetic friction :
  $\phi_k = \tan^{-1}\mu_k$

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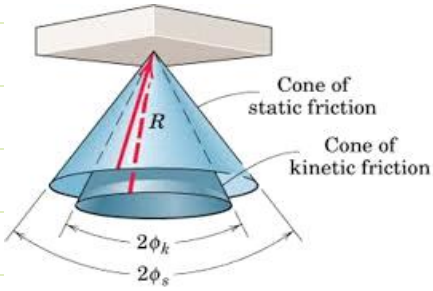
2. Cone of friction: The cone formed by Normal as height & friction force as radius is known as Cone of friction.

Every possible resultant of Normal & friction is present within the cone of static friction.

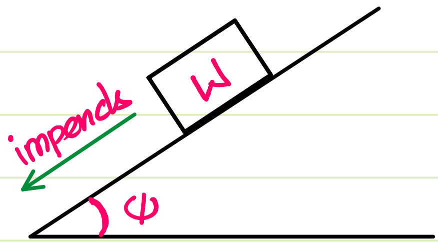


3. Angle of Repose: {ψ} The angle of inclination at which a body impends to slide down an incline is known as Angle of Repose.

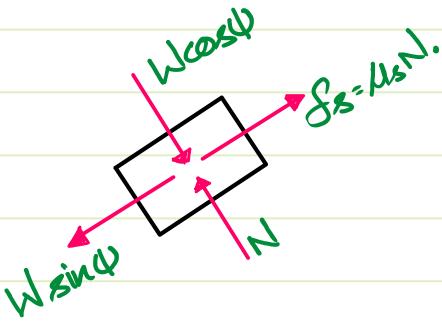
When a body impends motion, that is the last point of eq^m.

Note: In case of particle body

$\boxed{\psi = \phi_s}$



Proof:



By eq^m: $\Sigma F_{\perp \text{ar to incline}} = 0$

$\Rightarrow N = W\cos\psi$

$\Rightarrow f_s = \mu_s . W . \cos\psi$.

$\Sigma F_{\text{along incline}} = 0$

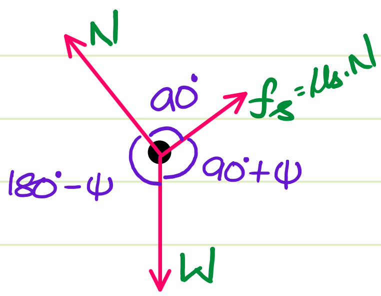
$f_s = W\sin\psi$

$\Rightarrow \mu_s . W\cos\psi = W\sin\psi$

$\Rightarrow \mu_s = \tan\psi$ {We also know $\mu_s = \tan\phi_s$}

$\therefore \psi = \phi_s$.

Alternate Method.

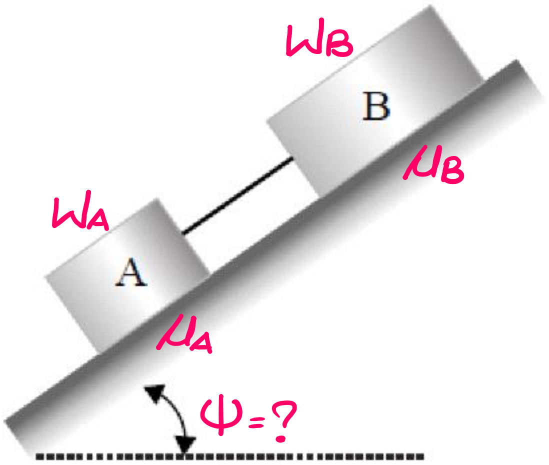


By Lamis:

$$\frac{\mu_s . N}{\sin\{180° - \psi\}} = \frac{N}{\sin\{90° + \psi\}} \Rightarrow \mu_s = \tan\psi$$

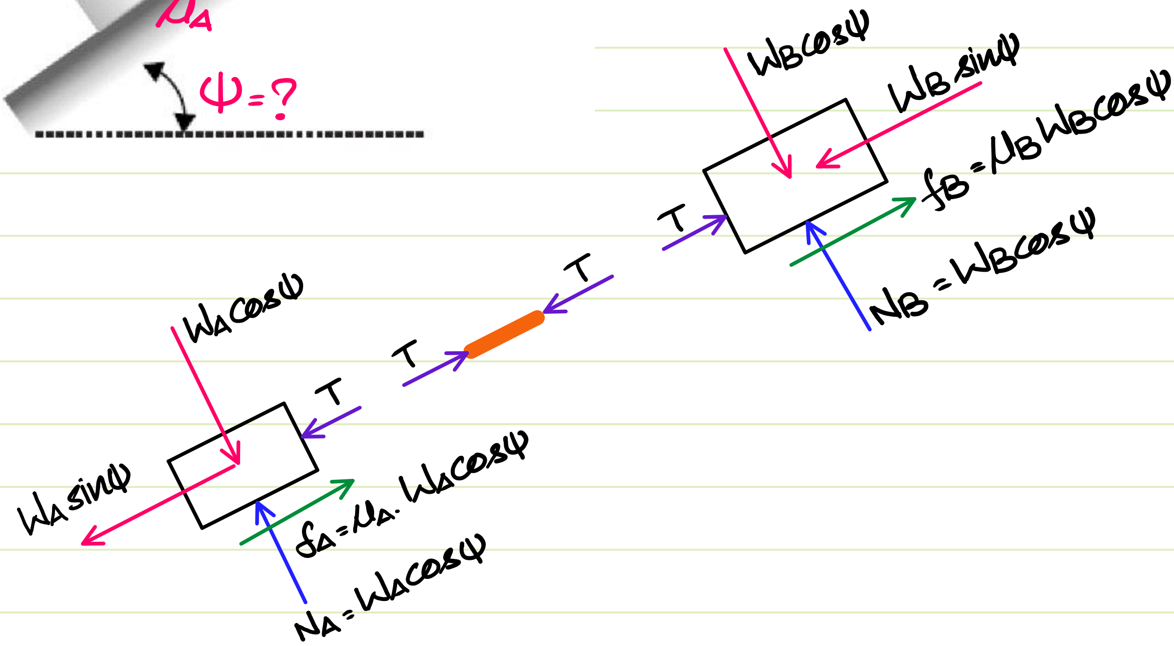
$\Rightarrow \tan\phi_s = \tan\psi$

$\psi = \phi_s$.



Given: Two particle bodies $\{W_A, W_B\}$ connected by Massless rigid link

Find: The angle of repose for the assembly.



ΣF along the Incline for the entire system $= 0$

$$\Rightarrow f_A + f_B = W_A\sin\psi + W_B\sin\psi$$

$$\Rightarrow \cos\psi\{\mu_A W_A + \mu_B W_B\} = \{W_A + W_B\}\sin\psi$$

$$\Rightarrow \tan\psi = \left\{\frac{\mu_A W_A + \mu_B W_B}{W_A + W_B}\right\}$$

$$\Rightarrow \boxed{\psi = \tan^{-1}\left\{\frac{\mu_A W_A + \mu_B W_B}{W_A + W_B}\right\}}$$

Note: If n blocks were connected, then

$$\psi = \tan^{-1}\left\{\frac{\sum_{i=1}^{n}(\mu W)_i}{\sum_{i=1}^{n} W_i}\right\}$$

Abdul Bari - 9573745717

1 Mark

Which among the following statements is true for a body moving on a dry surface under the action of applied forces?

(A) Kinetic-friction force is zero.

(B) Kinetic-friction force is equal to the static-friction force.

(C) Kinetic-friction force is greater than the static-friction force.

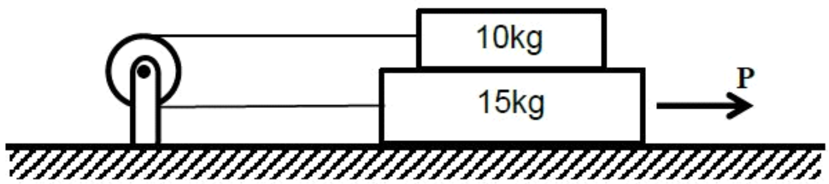
(D) ✓ Kinetic-friction force is lower than the static-friction force.

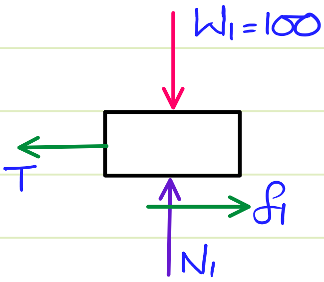
2 Marks.

Two rigid blocks, of masses **10kg** and **15kg**, are arranged one on top of the other and placed on a horizontal rough surface as shown. The blocks are connected to each other through an inextensible cable passing over a frictionless pulley. The coefficients of static friction between the blocks and also between the bottom block and the surface are all equal to **0.3**. The force ***P*** (in **Newtons**) needed to set the blocks in motion towards right is ________.

$\Rightarrow$ Eq^m is valid.

(Assume acceleration due to gravity **g = 10m/s²**).



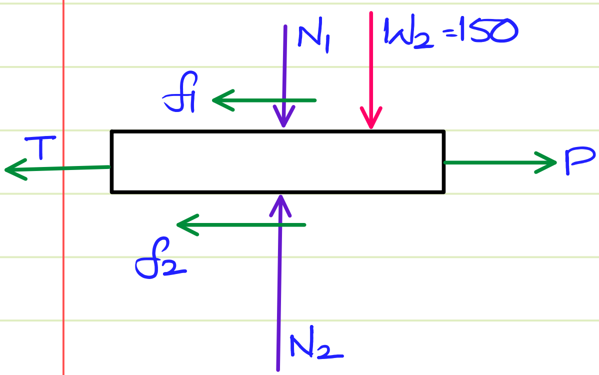


For the 10kg block:

$\Sigma F_y = 0 \Rightarrow N_1 = 100$

$f_1 = \mu N_1 = 30N.$

$\Sigma F_x = 0 \Rightarrow T = f_1 = 30N.$



For 15kg block:

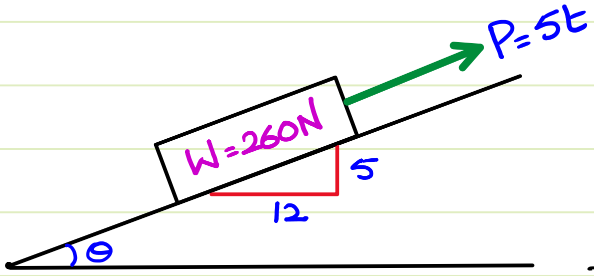
$\Sigma F_y = 0 \Rightarrow N_2 = N_1 + W_2$

$= 250N.$

$f_2 = \mu N_2 = 75N.$

$\Sigma F_x = 0 \Rightarrow P = f_1 + T + f_2$

$\therefore \boxed{P = 135N.}$

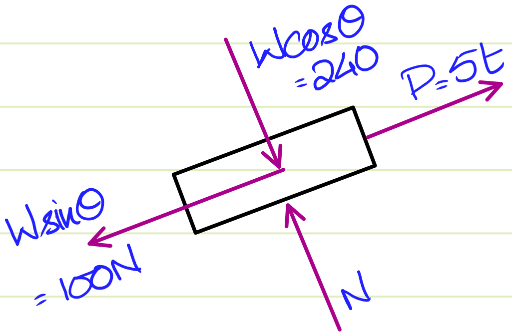


A force P=5t parallel to the incline acts as shown

$\mu_s = 0.5$; $\mu_k = 0.25$

find the values of friction at $t_1 = 10s$, $t_2 = 25s$, $t_3 = 50s$.

$\sin\theta = \frac{5}{13}$; $\cos\theta = \frac{12}{13}$



N = 240N

$f_s = 120N$ $f_k = 60N$.

* If the body demands any amount of force less than or equal to f_s in order to remain at rest, then that value of force is acceptable for st. friction

i) At $t_1 = 10s$; P = 50N.

demand for eq^m : $f = 50$

ii) At $t_2 = 25s$; P = 125N.

demand for eq^m : $f = 25$

iii) At $t_3 = 50s$; P = 250N.

demand for eq^m : $f = 150$ ✗

we have dyn cond^n ∴ $f = f_k = 60N$

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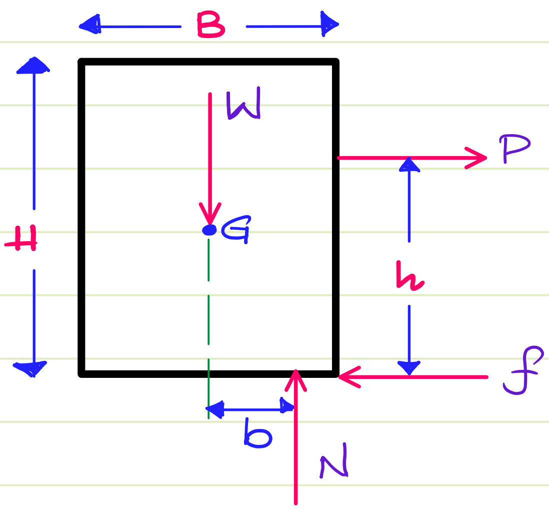
Case 2: Friction In Case of Rigid Body:

In case of rigid bodies, the forces acting may not be concurrent ⇒ There is a possibility of Moment (or) couple to be induced.

If the applied force & friction creates a disturbing couple then W & N try to create a restoring couple & for this to happen W & N must become eccentric w.r.t. each other.



$\Sigma F_x = 0$
$\Rightarrow P = f$
$\Sigma F_y = 0$
$\Rightarrow W = N$
$\Sigma M = 0$
$f h = W.b$

There are two possibilities due to these forces.

1. Toppling Effect: Normal reaches the corner about which toppling is to take place before friction could attain limiting static value.
2. Sliding Effect: Friction attains limiting static value $\{f_s\}$ before Normal could reach the point of toppling.

*** How to solve Numericals ?

Step1: Find f_s & the Max. possible eccentricity b/w W & N.

Step2: Find these 2 couples: $C_1 = C$ by P & f_s. {Max. dist. Coup}

$C_2 =$ Max C by W & N {Max rest. Couple}

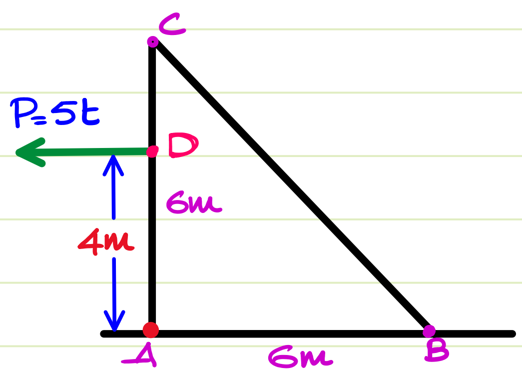
Abdul Bari - 9573745717

Step 3 : Compare C_1 & C_2

a. Couple by P & f_s < Max Couple by W & N. ⇒ {Sliding}

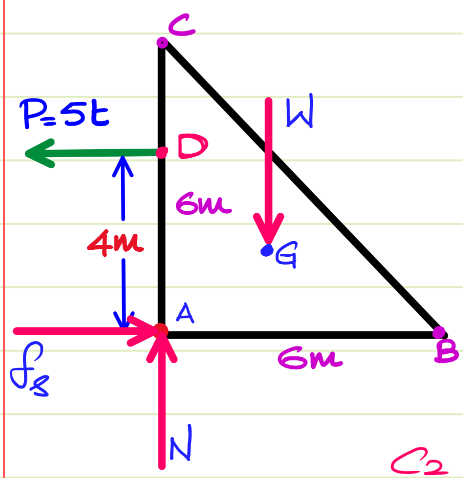
b. Max Couple by W & N < Couple by P & f_s. ⇒ {Toppling}

2 Marks.



$W = 200N$; $\mu_s = 0.6$

find the time at which Motion impends & also find the type of Motion that impends.



$f_s = \mu_s . N = 0.6 \times 200$ $\therefore f_s = 120N.$

Max possible ecc. b/w W & N = 2m.

C_1 : Max disturbing couple $= f_s \times 4$

$= 120 \times 4$

$\therefore C_1 = 480$ N.m.

C_2 : Max. restoring couple $= W \times 2$

$= 200 \times 2$

$\therefore C_2 = 400$ N.m.

$\because C_2 < C_1$ ∴ Toppling takes place.

When toppling is impending to take place, eq^m is valid.

$\therefore \Sigma M_A = 0$

$W\{2\} = 5t\{4\}$

$\Rightarrow t = 400/20$

$\therefore t = 20s.$ $P = f = 5 \times 20$

$= 100N. < f_s.$